



SPECIALIZED ICs

555 Timer Multivibrators • Voltage Regulator ICs • Analog Multiplier MPY634

A study of monostable, astable & bistable operation, linear and adjustable regulators, and four-quadrant analog multiplication

Agenda



555 Timer IC

Internal architecture and pin functions

01



Multivibrator Modes

Monostable, Astable & Bistable operation

02



Voltage Regulator ICs

Fixed (78xx/79xx) and adjustable (LM317) regulators

03



Regulated Supply Design

Key considerations for practical power supplies

04



Analog Multiplier MPY634

Four-quadrant multiplication IC and applications

05

What Is the 555 Timer?

- An 8-pin general-purpose timer IC used to generate accurate time delays or oscillations.
- Internally consists of two comparators, an SR flip-flop, a discharge transistor, and a resistive divider network.
- The resistive divider fixes comparator thresholds at $2/3 V_{CC}$ (upper) and $1/3 V_{CC}$ (lower).
- Can operate in three configurations: Monostable, Astable, and Bistable.

PIN FUNCTIONS (8-PIN DIP)

- 1 Ground
- 2 Trigger
- 3 Output
- 4 Reset (active low)
- 5 Control Voltage
- 6 Threshold
- 7 Discharge
- 8 VCC (+5V to +15V)

Monostable Multivibrator



One-Shot Operation

- Has ONE stable state (output LOW); a trigger pulse produces a single output pulse of fixed duration.
- A negative trigger pulse at pin 2 (below $1/3 V_{CC}$) sets the flip-flop; output goes HIGH and capacitor C charges through R.
- When capacitor voltage reaches $2/3 V_{CC}$, the threshold comparator resets the flip-flop and output returns LOW.
- Pulse width: $t = 1.1 R C$
- Applications: timers, pulse stretching, frequency dividers, switch debouncing.

TIMING DIAGRAM

Trigger (Pin 2)



Output (Pin 3)



Capacitor Voltage (Pin 6)



$2/3 V_{CC}$ threshold

Output pulse width depends only on R and C — not on the trigger pulse width.

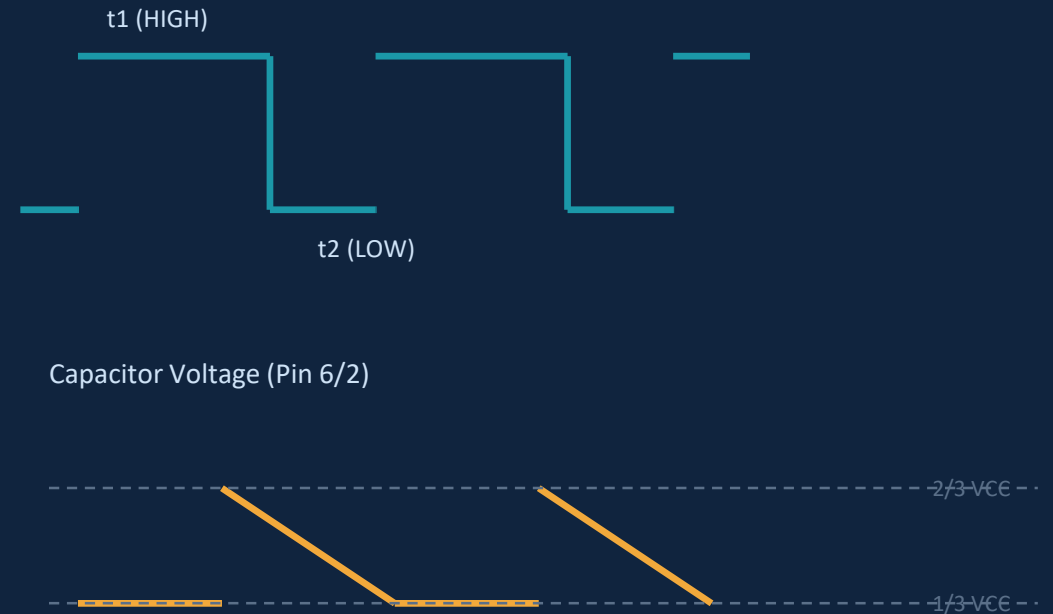
Astable Multivibrator



Free-Running Oscillator

- Has NO stable state — the output continuously switches between HIGH and LOW, generating a rectangular wave.
- Capacitor C charges through RA + RB toward 2/3 VCC, then discharges through RB alone toward 1/3 VCC — repeating indefinitely.
- Charge time (output HIGH): $t_1 = 0.693 (RA + RB) C$
- Discharge time (output LOW): $t_2 = 0.693 RB C$
- Frequency: $f = 1.44 / [(RA + 2RB) C]$
- Applications: clock/pulse generators, LED flashers, tone generators, PWM.

OUTPUT WAVEFORM



Duty cycle > 50% since charging (RA+RB) is always slower than discharging (RB).

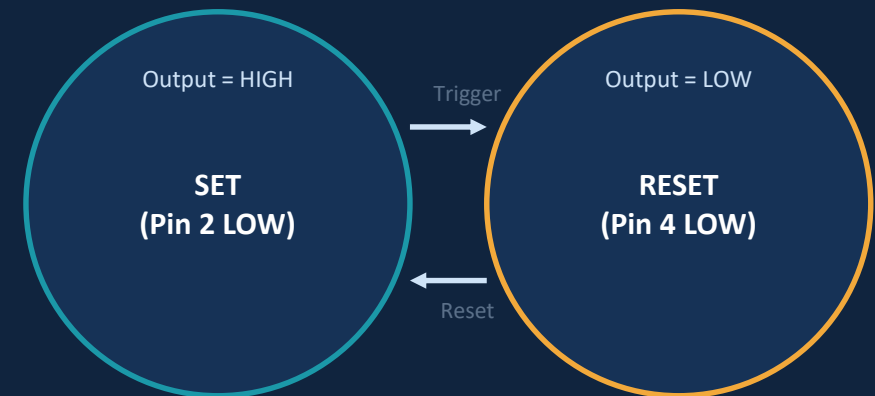
Bistable Multivibrator



Free-Running Flip-Flop (No Timing Components)

- Configured as a simple SET-RESET latch — has TWO stable states and no capacitor is required for timing.
- Pin 2 (Trigger) acts as SET: a LOW pulse drives output HIGH.
- Pin 4 (Reset) acts as RESET: a LOW pulse drives output LOW.
- Output remains in whichever state it was last driven to — it changes only on command, never on its own.
- Applications: manual switch debouncing, latching alarms, ON/OFF control circuits.

TWO STABLE STATES



Since threshold (pin 6) is tied off or disabled, the timing comparator never fires — so there is no automatic transition, only external SET/RESET commands.

Three Modes at a Glance

Mode	Stable States	Timing Components	Output Behaviour
Monostable	One	R, C	Single pulse per trigger; width = $1.1RC$
Astable	None	RA, RB, C	Continuous rectangular wave; $f = 1.44/[(RA+2RB)C]$
Bistable	Two	None	Latches HIGH/LOW on SET/RESET commands



All three modes rely on the same internal comparators (thresholds at 1/3 VCC and 2/3 VCC) — only the external R-C network and connections at pins 2, 4, and 6 change the mode of operation.

Three-Terminal Regulator ICs

- Integrated circuits that provide a stable, regulated DC output from just three pins: Input, Ground/Adjust, and Output.
- Include internal reference, error amplifier, pass transistor, and protection circuitry — reducing external parts to a couple of capacitors.
- Two broad families:
 - Fixed-voltage regulators — output preset at the factory (e.g., 78xx / 79xx series)
 - Adjustable-voltage regulators — output set by the designer using external resistors (e.g., LM317 / LM337)

BLOCK DIAGRAM



Line regulation: output remains steady despite changes in input voltage

Load regulation: output remains steady despite changes in load current

FIXED VOLTAGE REGULATORS

78xx / 79xx Series



78xx Series — Positive Regulators

- Provides fixed POSITIVE output voltages (e.g. 7805 → +5V, 7812 → +12V).
- Last two digits of part number = output voltage.



79xx Series — Negative Regulators

- Provides fixed NEGATIVE output voltages (e.g. 7905 → -5V, 7912 → -12V).
- Used for dual-supply ($\pm$) analog/op-amp circuits.

KEY FEATURES

Only 2 pins configurable

3-terminal package (TO-220); Input, Common/GND, Output

Built-in protection

Thermal shutdown, short-circuit & current-limit protection

Minimal external parts

Input/output capacitors (0.33 μ F / 0.1 μ F typical) only

Fixed output tolerance

Typically $\pm 4\%$ — cannot be adjusted by the user

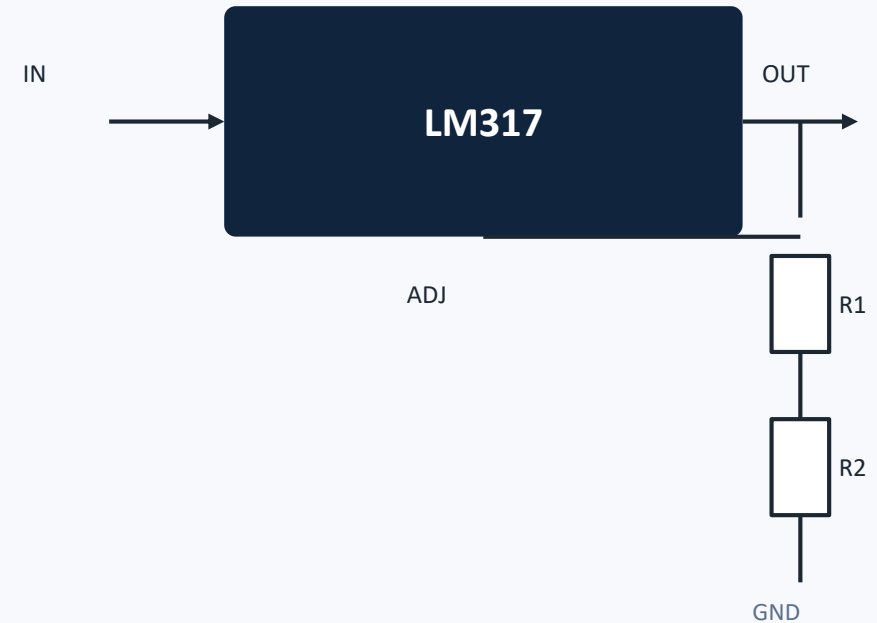
LM317 / LM337 — Adjustable Output

- Output voltage is set externally using a two-resistor divider (R1, R2), instead of being fixed at fabrication.
- Internal reference maintains a constant 1.25 V between the OUTPUT and ADJUST pins.

$$V_{out} = 1.25 \text{ V} \times (1 + R2/R1)$$

- LM317: positive adjustable, typically 1.25 V to 37 V.
- LM337: negative adjustable counterpart, -1.25 V to -37 V.
- Used where a variable bench supply or custom voltage rail is needed.

TYPICAL CONNECTION



R1 and R2 form a feedback divider that sets V_{out} via the ADJ pin.

DESIGN CONSIDERATIONS

Designing a Regulated Power Supply



DESIGN CONSIDERATIONS



Heat Dissipation

Power dissipated = $(V_{in} - V_{out}) \times I_{load}$; use adequate heatsinking to stay within safe junction temperature.



Dropout Voltage

V_{in} must exceed V_{out} by the regulator's minimum dropout (typically 2-3V) at all times, including ripple troughs.



Ripple & Filtering

Filter capacitor value must keep ripple low enough for the regulator's ripple-rejection to handle.



Load & Line Regulation

Choose a regulator with regulation specs suited to expected load-current swings and input variation.



Protection Features

Verify built-in thermal shutdown, current limiting, and safe-operating-area (SOA) protection.



Capacitor Placement

Place input/output capacitors close to IC pins per datasheet to ensure stability and prevent oscillation.

ANALOG MULTIPLIER

MPY634 Analog Multiplier IC

- A precision, wideband, four-quadrant analog multiplier IC manufactured by Texas Instruments (Burr-Brown).
- Produces an output proportional to the product of two differential input voltages X and Y .

$$W = [(X1-X2)(Y1-Y2)] / 10 V + Z2$$

- Four-quadrant: accepts positive AND negative values on both inputs.
- Internal scaling factor of $1/10V$ keeps output within a convenient $\pm 10V$ range.
- The Z input allows an offset or summing term to be added at the output.

FUNCTIONAL BLOCK



Two differential inputs (X , Y) are multiplied; Z provides feedback for divide/square-root configurations.

Applications of the Analog Multiplier

1

Multiplication

Direct product of two analog signals — $W = XY/10V$

2

Squaring

Tie X and Y inputs together: $W = X^2/10V$

3

Division

Use the Z-input feedback path to compute Z/X ratios

4

Square Root

Feedback configuration extracts $\sqrt{X}$ using the squarer in a loop

5

RMS-to-DC Conversion

Combine squaring and averaging to compute true RMS value

6

Modulation

Amplitude modulation, phase detection, and frequency doubling

SUMMARY

Key Takeaways



555 Timer

One versatile IC → three modes (monostable, astable, bistable) via external R-C and pin wiring



Regulator ICs

Fixed (78xx/79xx) for preset rails; adjustable (LM317/LM337) for custom, variable outputs



Power Supply Design

Balance heat dissipation, dropout margin, ripple filtering, and protection features



MPY634

Four-quadrant analog multiplier enabling multiply, square, divide, and RMS operations